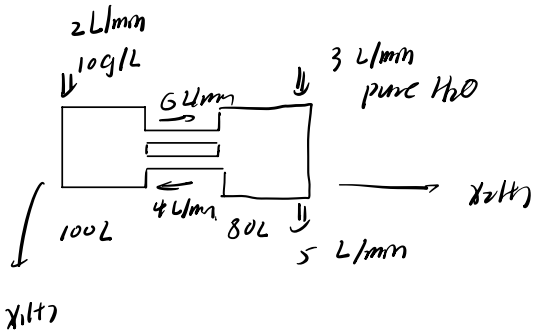
systems of 1st-order



$$x_1' = 20 + 4 \cdot \frac{x_2}{80} - 6 \cdot \frac{x_1}{100}$$

$$x_2' = 6 \cdot \frac{x_1}{100} - 9 \cdot \frac{x_2}{80}$$

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{3}{50} & \frac{1}{20} \\ \frac{3}{50} & -\frac{9}{80} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$v' = b + Av$$

$$v' - Av = b$$

$x'(t) = A(t)\,x(t) + q(t)$

where,

$x(t), q(t) \in \mathbb{R}^n$, $A(t) \in \mathbb{R}^{n\times n}$

Homogeneous: $q(t) = 0 \rightarrow x'(t) = A(t)\,x(t)$ Nonhomogeneous: $q(t) \neq 0$

eigenvalue

$$\begin{cases} \lambda_1 \neq \lambda_2 \neq \cdots \lambda_n \Rightarrow \sum_{i=1}^{n} c_i e^{\lambda_i t} v_i \\ \lambda = \alpha \pm i\beta \Rightarrow e^{\alpha t}(p\cos(\beta t) - q\sin(\beta t)),\ e^{\alpha t}(p\sin(\beta t) + q\cos(\beta t)) \end{cases} \quad \begin{cases} p = \mathrm{Re}(v) \\ q = \mathrm{Im}(v) \end{cases}$$

- Fundemental matrix

A matrix $\Phi(t) \in \mathbb{R}^{n\times n}$ is a fundemental matrix of the homogeneous system $x' = A(t)x$

$\Phi'(t) = A(t)\cdot\Phi(t)$. $\wedge$ $\det(\Phi(t)) \neq 0$

$\Phi(t) = e^{At}$

$x_h(t) = \Phi(t)\cdot c$ (homogeneous general solution)

- variation of parameter (non-)

$x_h = \Phi(t)\cdot c$

$x_p(t) = \Phi(t)\cdot M(t)$

$$\begin{cases} x_p' = \Phi' M + \Phi M' = A\Phi M + \Phi M' = \Phi(AM + M') & ① \\ x_p' = Ax + q = A(\Phi\cdot M) + q & ② \end{cases}$$

① = ②, st $q = \Phi M' \Rightarrow M' = \Phi^{-1} q$

$M = \int M'\,dt = \int \Phi^{-1} q\,dt$

st $x(t) = \Phi(t)\cdot\left(c + \int \Phi(t)^{-1} q(t)\,dt\right)$

Q1 $x' = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} x$, $x = \begin{pmatrix} x \\ y \end{pmatrix}$

$\det = (A - I\lambda) = \det \begin{pmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{pmatrix} = (2-\lambda)(3-\lambda) = 0 \Rightarrow \lambda = 2, 3.$

$A - I\lambda = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$x = c_1 e^{2t} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \cdot e^{3t} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q2: $x' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} x$

$\det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 + 1 = 0$, $\lambda = \pm i$ $\quad e^t \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \quad e^t \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$,

$\lambda = i \Rightarrow \begin{pmatrix} i \\ 1 \end{pmatrix} = \underset{p}{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} + \underset{q}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}} i \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \sin t \\ \cos t \end{pmatrix}$

$x_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cos t - \begin{pmatrix} 1 \\ 0 \end{pmatrix} \sin t$, $\quad x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \sin t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos t$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos t & \sin t \\ \sin t & -\cos t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$x = c_1 x_1 + c_2 x_2 = \begin{pmatrix} c_2 \\ c_1 \end{pmatrix} \cos t + \begin{pmatrix} -c_1 \\ c_2 \end{pmatrix} \sin t = \begin{pmatrix} c_2 \cos t + c_1 \sin t \\ +c_2 \sin t - c_1 \cos t \end{pmatrix}$

Q3: $x' = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x$

$\lambda = 1$.

$A - I\lambda = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $\quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \left(\begin{array}{cc|c} 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right) \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$x_1 = e^t v_1$ 最高次系数1

$x_2 = e^t (t v_1 + v_2)$

$x(t) = c_1 x_1 + c_2 x_2 = c_1 e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^t \begin{pmatrix} t \\ 1 \end{pmatrix}$

Q4: $x' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} x + \begin{pmatrix} e^t \\ 1 \end{pmatrix}$

$$x(t) = e^{At} c + \int_0^t e^{A(t-s)} q(s)\, ds$$

$$e^{At} = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \Rightarrow e^{A(t-s)} = \begin{pmatrix} e^{t-s} & 0 \\ 0 & e^{2(t-s)} \end{pmatrix}$$

$$\int_0^t e^{A(t-s)} \cdot q(s)\, ds = \int_0^t \begin{pmatrix} e^{t-s} & 0 \\ 0 & e^{2(t-s)} \end{pmatrix} \begin{pmatrix} e^s \\ 1 \end{pmatrix} ds$$

$$= \int_0^t \begin{pmatrix} e^t \\ e^{2(t-s)} \end{pmatrix} ds = \begin{pmatrix} \int_0^t e^t\, ds \\ \int_0^t e^{2(t-s)}\, ds \end{pmatrix} = \begin{pmatrix} te^t \\ \frac{e^{2t}-1}{2} \end{pmatrix}$$

homogeneous solution

$$e^{At} c = \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix}$$

$$e^{At} = P e^{Dt} P^{-1}$$

$$\det\begin{pmatrix} a-r & b \\ c & d-r \end{pmatrix} = r^2-(a+d)+(ad-bc) = r^2 - tr(A)\cdot r + \det(A)$$

eigenvector $r = \lambda + iM$

complex eigenvector $\begin{pmatrix} a \\ b \end{pmatrix} + i\begin{pmatrix} c \\ d \end{pmatrix}$

$$y = e^{\lambda t}\begin{pmatrix} a & c \\ b & d \end{pmatrix}\begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$\lambda_1 = \lambda_2 \to v$

$y_1 = e^{\lambda t} v$, $y_2 = te^{\lambda t} v + Me^{\lambda t}$, where $(A-\lambda I)M = v$

e.g. $\vec{x}' = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix}\vec{x}$

$tr(A) = -4$, $\det(A) = 3-(-1)(1) = 4$

$\Rightarrow \lambda^2 + 4\lambda + 4 = 0$

$\lambda = -2$

$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \to v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\vec{x}_1 = e^{-2t}\cdot\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\left(\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 0 & 0 \end{array}\right)$

$\vec{x}_2 = te^{-2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + e^{-2t}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$c_1 y_1 + c_2 y_2 = c_1 e^{-2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 t e^{-2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= c_1 e^{-2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t}\begin{pmatrix} t+1 \\ t \end{pmatrix}$$

Non-homogenous system

$$\vec{x}' = Ax + g$$

where $g = g(t)$

1° variation of parameter

2° match coefficients

Recall: Given that a fundamental set $\{y_1, y_2\}$ of $y_1'' + py_1' + qy = 0$ construct

$$y = u_1 y_1 + u_2 y_2 = \underbrace{[y_1\ y_2]}_{\text{complementary solution}} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

integrating factor

$$y_p = \underbrace{[\vec{x}_1, \vec{x}_2]}_{2\times 2} \underbrace{\vec{u}}_{2\times 1} = X\vec{u}$$

sub $\vec{x} = X\vec{u}$, $\vec{x}' = A\vec{x} + g \Rightarrow \frac{d}{dt}(X\vec{u}) = AX\vec{u} + \vec{g}$

$$\vec{x}' = \underbrace{(\vec{x}_1', \vec{x}_2')}_{A\vec{x}_1,\ A\vec{x}_2}\vec{u} + [x_1\ x_2]\vec{u}' = AX\vec{u} + X\vec{u}'$$

$$\Rightarrow \cancel{AX\vec{u}} + X\vec{u}' = \cancel{AX\vec{u}} + \vec{g}$$

$$\vec{u}' = X^{-1}\vec{g}$$

$$\vec{u} = \int X^{-1}\vec{g}\, dt$$

e.g.

$$\vec{x}' = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}\vec{x} + \begin{pmatrix} e^t \\ -e^t \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix} \to \lambda^2 - \lambda - 6 = (\lambda-3)(\lambda+2) = 0 \Rightarrow \lambda = 3, -2$$

$$\vec{x}_1 = e^{3t}\begin{pmatrix}1\\1\end{pmatrix}, \quad \vec{x}_2 = e^{-2t}\begin{pmatrix}1\\-4\end{pmatrix}$$

$$X = \begin{pmatrix} e^{3t} & e^{-2t} \\ e^{3t} & -4e^{-2t} \end{pmatrix} \qquad \det(X) = -4e^{t} - e^{t} = -5e^{t}$$

$$X^{-1} = -\frac{1}{5}e^{-t}\begin{pmatrix} -4e^{-2t} & -e^{-2t} \\ -e^{3t} & e^{3t} \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 4e^{-3t} & e^{-3t} \\ e^{2t} & -e^{2t} \end{pmatrix}$$

$$\vec{M} = \int X^{-1}\vec{g}\,dt = \frac{1}{5}\int \begin{pmatrix} 4e^{-3t} & e^{-3t} \\ e^{2t} & -e^{2t} \end{pmatrix}\begin{pmatrix} e^{t} \\ -e^{t} \end{pmatrix} dt$$

$$= \frac{1}{5}\int \begin{pmatrix} 3e^{-2t} \\ 2e^{3t} \end{pmatrix} dt = \frac{1}{5}\begin{pmatrix} -\frac{3}{2}e^{-2t} + c_1 \\ \frac{2}{3}e^{3t} + c_2 \end{pmatrix}$$

$$y_p = X\vec{M} = \begin{pmatrix} e^{3t} & e^{-2t} \\ e^{3t} & -4e^{-2t} \end{pmatrix} \frac{1}{5}\begin{pmatrix} -\frac{3}{2}e^{-2t} \\ \frac{2}{3}e^{3t} \end{pmatrix}$$

$$= \frac{1}{5}\begin{pmatrix} -\frac{3}{2}e^{t} + \frac{2}{3}e^{t} \\ -\frac{3}{2}e^{t} - \frac{8}{3}e^{t} \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -\frac{5}{6}e^{t} \\ -\frac{25}{6}e^{t} \end{pmatrix}$$

$$= \frac{1}{5}\cdot\left(-\frac{5}{6}\right)e^{t}\begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\cancel{-45-4}$$
$$\cancel{-45+22}$$
$$\cancel{3\times 5-10<}$$

$$= -\frac{e^t}{6}\begin{pmatrix}1\\5\end{pmatrix}$$

Exam 2:

$$\vec{x}' = \begin{pmatrix}2 & -3\\1 & -2\end{pmatrix}\vec{x} + \begin{pmatrix}2\\1\end{pmatrix}t$$

$$\lambda^2 - 1 = 0$$

$$\lambda = 1, -1$$

$$x_p = \vec{a}t + \vec{b}$$

$$x_p' = \vec{a}$$

$$\vec{a} = A(\vec{a}t + \vec{b}) + \begin{pmatrix}2\\1\end{pmatrix}t$$

$$\begin{cases} A\vec{a} + \begin{pmatrix}2\\1\end{pmatrix} = 0 \quad (1) \\ \vec{a} = A\vec{b} \end{cases}$$

$$b = \begin{pmatrix}-1\\-1\end{pmatrix}$$

$$-2 \quad 4 \quad 0$$

$$\left(\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\begin{pmatrix}-2\\-1\end{pmatrix}\right)$$

$$2 \quad -2$$

$$1 \quad 0 \quad -1$$
$$0 \quad 1 \quad 0$$

$$a = \begin{pmatrix}-1\\0\end{pmatrix}$$

$$-2 \quad 4 \quad 2$$

st

$$x_p = t\begin{pmatrix}-1\\0\end{pmatrix} - \begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}-t-2\\-1\end{pmatrix}$$

$$\lambda = \pm i$$

$$\lambda^2 + 1 = 0$$

$$x_p = \vec{a}\,\underline{\sec t} + \vec{b}$$

$$A = \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}$$

$$x_p' = \sec t \cdot \tan t\,\vec{a}$$

$$\cancel{\sec t \cdot \tan t \cdot \vec{a}} = A\vec{a}\sec t + A\vec{b} + \cancel{\begin{pmatrix}2\\3\end{pmatrix}\sec t}$$

$$A\vec{a}\sec$$

$$\begin{cases} \tan t \cdot \vec{a} = \begin{pmatrix}2\\3\end{pmatrix} \\ A(\underbrace{\sec t\,\vec{a} + \vec{b}}) = 0 \end{cases}$$

$$\vec{a} = \begin{pmatrix}2\\3\end{pmatrix}\cot$$

$$y_p = \Phi(t)\cdot\vec{M} \qquad A\Phi = \Phi'$$

$$y_p' = \cancel{A\Phi(t)\vec{M}} + \Phi(t)\cdot\vec{M}' = \cancel{Ay_p} + g$$

$$M' = \int \Phi^{-1}(t)\,\vec{g}\,dt$$

$$= \int \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \cdot \begin{pmatrix} 2\sec t \\ 3\sec t \end{pmatrix} dt$$

$$= \int \begin{pmatrix} 2 - 3\tan t \\ 2\tan t + 3 \end{pmatrix} dt$$

$$M = \begin{pmatrix} 2t - 3\ln|\sec t| \\ 2\ln|\sec t| + 3t \end{pmatrix}$$

$$y_p = \Phi M = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \begin{pmatrix} 2t + 3\ln|\cos t| \\ -2\ln|\cos t| + 3t \end{pmatrix}$$

$$= \boxed{\begin{matrix} 2t\cos t + 3\cos t\ln|\cos t| - 2\sin t\ln|\cos t| + 3\sin t\, t \\ -2t\sin t - 3\sin t\ln|\cos t| - 2\cos t\ln|\cos t| + 3t\cos t \end{matrix}}$$

$$\lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1) = 0$$

$$\lambda = 3,\ 1$$

$$\begin{pmatrix} 0 & 0 \\ -1 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & -2 \\ 0 & 0 \end{pmatrix} \rightsquigarrow v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \qquad \lambda = 1$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \rightsquigarrow v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \lambda = 3$$

$$\Phi = \begin{pmatrix} 2e^{t} & 0 \\ e^{t} & e^{3t} \end{pmatrix} \qquad \det(\Phi) = 2e^{4t}$$

$$\Phi^{-1} = \frac{1}{2} \cdot \begin{pmatrix} e^{3t} & 0 \\ -e^{t} & 2e^{t} \end{pmatrix} \cdot e^{-4t}$$

$$= \frac{1}{2} \begin{pmatrix} e^{-t} & 0 \\ -e^{-3t} & 2e^{-3t} \end{pmatrix}$$

$$\Phi^{-1}\vec{g} = \frac{1}{2} \begin{pmatrix} e^{-t} & 0 \\ -e^{-3t} & 2e^{-3t} \end{pmatrix} \begin{pmatrix} e^{2t} \\ -e^{2t} \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} e^{t} \\ -3e^{-t} \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} e^{t} \\ -3e^{-t} \end{pmatrix}$$

$$\frac{1}{2} \left[\begin{pmatrix} e^{t} \\ -e^{-t} \end{pmatrix}^{T} \begin{pmatrix} 0 \\ -2e^{-t} \end{pmatrix} \right]$$

$$x_p(t) = \Phi \int \Phi^{-1}(t)\, \vec{g}\, dt$$

$$= \begin{pmatrix} 2e^{t} & 0 \\ e^{t} & e^{3t} \end{pmatrix} \int \frac{1}{2} \begin{pmatrix} e^{t} \\ -3e^{-t} \end{pmatrix} dt$$

$$= \frac{1}{2}\begin{pmatrix} 2e^{t} & 0 \\ e^{t} & e^{3t} \end{pmatrix}\begin{pmatrix} e^{t} \\ 3e^{-t} \end{pmatrix}$$

$$= \frac{1}{2}\left[\begin{pmatrix} 2e^{2t} \\ e^{2t} \end{pmatrix} + \begin{pmatrix} 0 \\ 3e^{2t} \end{pmatrix}\right] = \begin{pmatrix} e^{2t} \\ 2e^{2t} \end{pmatrix}$$

$$x(t) = \begin{pmatrix} e^{2t} \\ 2e^{2t} \end{pmatrix} + c_1\begin{pmatrix} 2e^{t} \\ e^{t} \end{pmatrix} + c_2\begin{pmatrix} 0 \\ e^{3t} \end{pmatrix}$$

$$x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_1\begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$$\begin{cases} 2c_1 = 1 \\ c_1 + c_2 = 3 \end{cases}$$

$$\Rightarrow c_1 = \frac{1}{2}, \quad c_2 = \frac{5}{2}$$